On the Redistribution of Governance Friction in Integrated Data Ecosystems

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April 1, 2026

Abstract

Introduction

The binding constraint on Integrated Data System (IDS) performance is often *governance adequacy*: the capacity of coordination boundaries to sustain legitimate, structurally-appropriate authorizations as partnerships expand across heterogeneous domains and institutional layers.

Objectives

We investigate how governance saturation in IDSs can be diagnosed from routine operational artifacts before it manifests as project failure.

Methods

Rather than starting from a predetermined theoretical framework, we develop a set of minimal, episode-level governance indicators that are reconstructible from routine operational artifacts. We compute these and derived indicators to retrospectively diagnose governance delays and stalls in nine IDS projects comprising 30 governance episodes across three ecosystems, three domains (Education, Justice, Health), and three institutional layers (local, regional, state), covering approximately 600,000 individuals.

Results

1. All six stalled episodes occur within custodial projects, while federated projects exhibit zero stalls across 17 episodes despite higher mean coupling loads (7.16 vs. 4.50).
2. Mean normalized time-to-value burden is 78.65 in custodial projects versus 46.34 in federated projects; within the federated projects, steward-present projects achieve 19.36 versus 59.83 without stewards.
3. The highest and lowest project time-to-value in the dataset are both in a single ecosystem: the custodial project P1 reaches 624 days, while the steward-present federated project P3 achieves 87 days at the highest coupling load in the dataset (16.90). The two steward-present projects handle the largest partner networks ($n_p =$

9 and 7) at high coupling loads (16.90 and 7.00), yet low normalized burdens among their respective strata (5.15 and 33.57).

Conclusions

Governance coordination in Integrated Data Systems is often framed as a problem of shared understanding, yet existing models that focus on single dimensions of governance adequacy struggle to diagnose delays in real projects.

Scalable governance designs should address both structural resolution and functional capacity and consider structural variations such as modularization, Privacy-Enhancing Technologies (PETs), and domain stewardship. The most important design question is not whether governance friction can be eliminated, but whether it is allocated to boundaries with appropriate resolution and capacity. These findings have direct implications for the design of composable, federated data architectures that are increasingly adopted in Integrated Data Systems.

1 Introduction

Integrated Data Systems (IDSs) link administrative records across organizations in a variety of domains, such as health, education, housing, and justice, in order to support evidence-based decision-making.

Operating an IDS requires aligning three interdependent concerns: governance (authority, accountability, legitimacy), infrastructure (identity-handling, controls, access, auditability), and culture (risk language, decision cadence, funding, political and organizational realities).

Of the three concerns, the binding constraint on system performance is often *governance adequacy*: the capacity of coordination boundaries to sustain legitimate, structurally-appropriate authorizations as partnerships expand across heterogeneous domains and institutional layers. In these scenarios the number of cross-boundary decisions requiring reconciliation often grows faster than the capacity of centralized governance bodies to issue legitimate, structurally-appropriate authorizations in a timely manner – their maximum *governance throughput*. *Governance saturation* is the condition in which the volume, speed, or complexity of governance demands exceeds a system maximum governance throughput, causing delays, bottlenecks, or loss of effectiveness.

We investigate how governance saturation in IDSs can be diagnosed from routine operational artifacts before it manifests as project failure. Rather than starting from a predetermined theoretical framework, we identify minimal observability indicators, apply them retrospectively to IDS implementations spanning centralized custody and federated designs, and develop a new model from the patterns that emerge.

1.1 Problem framing

The problem of governance saturation can be understood in terms of governance boundaries, boundary configurations, and the friction associated with authorization decisions.

A *governance boundary* is an institutional mechanism—a committee, a review board, an agreement process—that makes authorization decisions for a given IDS. In practice, authorization requests are always mediated by one or more individuals who represent the institution to the IDS ecosystem and who bear designated roles within the institution, such as data steward, privacy officer, or governance lead. The perspective, priorities, and finite discriminative capacity that such a role-bearer brings to the governance request mediation process collectively constitute what we term the boundary's *configuration*.

Governance friction is the resistance that arises when processing an authorization request in an IDS. Governance friction may be minimal, allowing requests to glide through an authorization process with little resistance. In other cases, resistance may be significant and stem from the essential socio-technical complexity of processing integrated data in an IDS.

Governance friction is not inherently pathological; some governance friction reflects appropriate deliberation. The diagnostic question is whether the observed friction is proportionate to the complexity of the request, or whether the friction reflects limitations in how

the boundary itself is organized.

1.1.1 Structural differences

Governance coordination in IDSs is often framed as a problem of shared understanding: if the partners involved can agree on definitions of privacy, compliance, and trust, the reasoning goes, governance will follow. But shared understanding is constrained by a deeper structure.

Consider an IDS that begins by linking education records across school districts within a single county. The governance boundary—say, a data governance committee—develops review procedures calibrated to this scope: a common statutory regime (FERPA [1]), familiar institutional actors, and a well-understood set of data elements.

When the same IDS later seeks to incorporate health records (governed by HIPAA [2]) or justice records (governed by state statute and court orders), the committee faces authorization requests that require fundamentally different evaluation criteria, different legal expertise, and different accountability structures (Table 1).

Table 1: **IDS governance activities by functional area.**

Functional area	**Activities**	**Input → Evaluator → Outcome**
IDS Coordination	Governance formation	Charter → voting partners → approve/deny
	Participant onboarding	Request + role → evaluators → approve/deny
	Data sharing agreement	Security posture → agreement evaluators → approve/deny
	Product feasibility	Assessment → feasibility evaluators → approve/deny
Contributor Interface	Interface spec. update	Existing specs → change evaluators → publish/deny
Data Product	Data use license	Use, elements, method → signatory → approve/deny
	Data pull request	Request → DSA + contributor → approve/deny
	Product build/release	Composite facesheet → data + contributor → release/cancel

As a practical example, consider the term "deidentified data". This term carries different structural implications for health and education data (Table 2): under HIPAA, proper deidentification can remove data from the regulation's scope entirely; under FERPA, it cannot. Both regimes aim to protect individual privacy, but they operationalize protection through different mechanisms. A governance boundary that treats "deidentified data" under FERPA

and "deidentified data" under HIPAA as equivalent will misroute authorizations regardless of how well the partners understand each other. We call this condition *conflation*. Under conflation, requests that require different treatment receive identical processing, producing delays, stalls, or misrouted authorizations regardless of the competence of the people involved.

Governance designs must be sensitive to the degree to which incompatible request types are concentrated at a single governance boundary. By the same token, diagnostic frameworks such as the one presented here must account for these incompatibilities. Structured Incompleteness Theory (SIT) [3] helps us understand why: when faced with more structurally distinct request types than it can process in distinct ways, a governance boundary must treat some of those distinct types identically. The boundary's configuration—which request types it can tell apart and which it conflates—limits governance throughput. When that configuration is too coarse for the requests the boundary faces, no amount of staffing or process improvement can produce governance that differentiates what the configuration conflates.

Table 2: **Structural differences between education (FERPA-regulated) and health (HIPAA-regulated) domains.** These reflect governance influences that a single boundary conflates at its peril.

Dimension	**Education (FERPA)**	**Health (HIPAA)**	**Governance implication**
Legitimacy basis	Educational purpose	Public health, treatment, payment	Different outcome criteria
Unit of protection	Education record	Protected health information	Different protected elements
Authorization trigger	School official exception	Permitted use categories	Non-isomorphic approval paths
Deidentification role	Does not remove FERPA scope	Can remove data from HIPAA scope	Different structural roles
Interpretive authority	Agency counsel	Privacy officer and regulators	No shared adjudicating body

1.2 Related work

This paper draws on and contributes to multiple literatures. In each case we identify a common limitation: existing work treats governance adequacy as a single dimension and provides no method for diagnosing governance inadequacy from the artifacts that governance processes already produce.

Organizational coordination theory. The foundational insight that organizations are information-processing systems whose design must match the uncertainty they face comes from Galbraith [4], building on Thompson's [5] typology of interdependence (pooled, sequential, reciprocal). Malone and Crowston [6] generalized this into a theory of coordination as the management of dependencies between activities. These frameworks explain *why* governance load grows with organizational complexity but treat the coordinating body's capacity as a single scalar: more capacity means better coordination. They do not distinguish between a boundary that lacks throughput and one that cannot differentiate among the requests it faces. Simon's [7] architecture of complexity and Baldwin and Clark's [8] modularity theory explain decomposition as the canonical response to coordination overload, but again characterize the problem as one of volume, not of structural mismatch between the boundary's configuration and the variety of requests it must handle.

Data governance. The data governance literature has matured from prescriptive frameworks toward richer accounts of institutional design. Khatri and Brown [9] identify five decision domains (data principles, quality, metadata, access, lifecycle) and argue that governance effectiveness depends on how decision rights are allocated. Abraham, Schneider, and vom Brocke [10] synthesize the field into a conceptual framework emphasizing structural, procedural, and relational mechanisms. Hardjono, Lipton, and Pentland [11] extend governance thinking to data trusts, where interoperability across institutional boundaries is a first-order design concern. These contributions clarify *what* governance must accomplish (legitimacy, accountability, decision-rights allocation) but treat the governance body's ability to produce adequate decisions as a function of institutional maturity and process quality rather than the ability to *structurally distinguish* among different types of request.

Privacy-enhancing technologies (PETs). A parallel literature shows that architectural choices can change the governance problem itself, not merely improve throughput within a fixed governance design. Secure multi-party computation [12] enables computation on distributed data without centralizing it. Differential privacy [13] provides formal guarantees that shift compliance from case-by-case adjudication toward verifiable properties of the computation. Each of these technologies can reduce the governance burden per query by replacing trust in custody with trust in verifiable controls. However, the PET literature focuses on the technical enforcement of privacy guarantees and largely treats governance as an external constraint rather than a system to be diagnosed and designed. The question of how architectural change redistributes—rather than eliminates—governance friction remains largely unexamined.

Composable and federated data architectures. The data mesh paradigm [14] reframes data infrastructure as a decentralized, domain-oriented network in which each domain owns and publishes its own analytical data products under federated governance. This

architectural philosophy aligns with the structural changes we observe in our federated stratum: modularization of data custody, domain-level stewardship, and governance boundaries calibrated to domain-specific compliance requirements. However, the data mesh literature focuses primarily on enterprise environments and does not address the cross-institutional, cross-statutory governance coordination that characterizes public sector Integrated Data Systems. Bean [15] adapts composable data architectures to the IDS context, arguing that centralized data warehouses create single points of governance failure and that federated, composable architectures redistribute governance responsibilities along domain lines. Our contribution complements this line of work by providing diagnostic tools for detecting governance friction that can be used to guide data architecture design.

IDS practice and the observability gap. There is a substantial practice literature emphasizing the institutional prerequisites for cross-agency data sharing using Integrated Data Systems. Frameworks for IDS development emphasize partnership building, legal analysis, and data quality [16, 17, 18], while empirical studies demonstrate the value of linked administrative data for social policy research [19, 20]. These sources provide rich case knowledge about what makes IDS partnerships succeed or fail. We extend this work by offering a systematic measurement of governance performance.

Requisite variety and structural mismatch. The closest theoretical ancestor to the structural dimension we identify is Ashby's Law of Requisite Variety [21]: a regulator must possess at least as much variety as the system it governs. Our work operationalizes this principle for governance boundaries, focusing on whether a boundary configuration draws sufficient distinctions among structurally different request types rather than the boundary's capacity per se.

1.3 Contributions

Existing work on data governance, PETs, and IDS practice does not provide methods for detecting governance inadequacy from the artifacts that governance processes already produce. Coordination theory diagnoses misfit at the organizational-design level; data governance prescribes decision-rights structures; PET research evaluates technical properties; IDS practice identifies success factors retrospectively.

The primary contribution of this paper is a four-layer analytical framework for governance observability, founded on minimal indicators reconstructed from routine operational artifacts. We apply this framework to the analysis of nine IDS implementation projects and identify four key patterns in the data. Finally, we argue that the observed patterns are not explainable by a one-dimensional model of governance adequacy, and develop a new, two-dimensional model from these patterns.

2 Methods

IDS governance work generates a trail of operational artifacts—agreements, approval emails, committee minutes, provisioning records—but these are rarely collected or analyzed systematically. We have developed an analytical framework that reconstructs episode-level governance indicators from these artifacts and preserves an auditable path from every derived metric back to at least one durable source document.

2.1 Analytical framework

The framework is organized as a four-layer hierarchy (Figure 1), moving from directly observed artifacts at the bottom to cross-ecosystem aggregation at the top. The lower layers (events, episodes) define what we observe; the upper layers (projects, ecosystems) define how we compare indicators within and between projects.

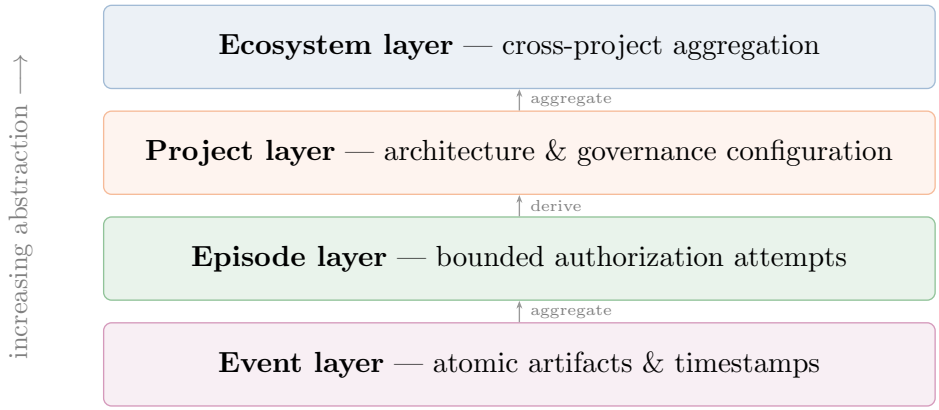

Figure 1: **Layered analytical framework.**

2.1.1 Event layer

An *event* is a timestamped occurrence in a governance process: a request submitted, an approval issued, an agreement executed, provisioning completed, an output delivered, or a withdrawal. Each event corresponds to a durable artifact—a signed agreement, an email, a committee minute entry, a provisioning log—from which the event's existence and timing can be reconstructed.

2.1.2 Episode layer

An *episode* aggregates a sequence of events into a bounded governance authorization attempt. Episodes are the primary units of our analysis.

Table 3 summarizes the indicators associated with episodes.

Table 3: **Episode-layer indicators.**

Indicator	Name	Definition
$t_{0,e}$	Episode start	Timestamp of earliest event indicating entry into an approval workflow
$t_{1,e}$	First durable authorization	Timestamp of earliest qualifying authorization event for core scope
$t_{2,e}$	First delivered value	Timestamp of earliest analytic output answering a stakeholder question
n_e	Partner count	Number of governance-relevant institutional actors
D_e	Domain set	Domains involved in an episode
L_e	Layer set	Institutional layers involved in an episode
$h_{d,e}$	Domain heterogeneity	Heterogeneity of domains involved in an episode
$h_{\ell,e}$	Layer heterogeneity	Heterogeneity of layers involved in an episode
$stall_e$	Pre-commit stall	Binary: 1 if no qualifying $t_{1,e}$ despite sustained engagement, otherwise 0
c_e	Coupling proxy	See below
$\tau_{1,e}$	Authorization latency	$t_{1,e} - t_{0,e}$
$\tau_{2,e}$	Time to delivered value	$t_{2,e} - t_{0,e}$
b_e	Normalized burden	Time to delivererd value per unit of coupling load: $\tau_{2,e}/c_e$

An event sequence qualifies as an episode if the sequence contains: (i) a recognizable initiation event at time $t_{0,e}$, (ii) evidence of governance engagement, and (iii) either a durable authorization (time $t_{1,e}$) or sustained engagement without authorization (stall). For stalled episodes ($stall_e = 1$), authorization latency $\tau_{1,e}$ is undefined; these episodes are analyzed separately as a binary saturation indicator.

Each episode involves one or more *partners*: the institutional actors (agencies, departments, universities, nonprofits) whose participation requires governance authorization. Partners are characterized by their domain $d \in D_e$ (e.g., education, health, justice) and their institutional layer $\ell \in L_e$ (e.g., local, regional, state). From these counts we calculate the domain and layer heterogeneity of an episode using Simpson-style concentration indices:

$$h_{d,e} = 1 - \sum_{d \in D_e} \rho_{d,e}^2 \qquad h_{\ell,e} = 1 - \sum_{\ell \in L_e} \rho_{\ell,e}^2.$$

where $\rho_{d,e}$ is the proportion of partners in a domain d for a given episode e and $\rho_{\ell,e}$ is the proportion of partners in a layer l for a given episode e.

Finally, we derive the episode's coupling-load proxy c_e, which captures both scale (n_e,

the number of partners) and structural heterogeneity ($h_{d,e}$ and $h_{\ell,e}$):

$$c_e \;=\; n_e \times (1 + h_{d,e}) \times (1 + h_{\ell,e}).$$

2.1.3 Project layer

A project groups episodes that share a common set of partners, a common governance boundary design (e.g., the same committee or agreement structure), and a common data architecture (custodial or federated).

Table 4: **Project-layer indicators.**

Indicator	**Name**	**Definition**
$arch_p$	Architecture	$arch_p = 0$: custodial (raw data centralized). $arch_p = 1$: federated, PET-enabled
$stew_p$	Domain steward presence	$stew_p = 1$: authorized domain representative. $stew_p = 0$: no such role
E_p	Set of episodes in a project	
e_p^*	Delivery episode	The episode in E_p whose $t_{2,e}$ corresponds to the first delivered data product
$c_{e_p^*}$	Delivery coupling proxy	The coupling proxy specific to e_p^*
b_p	Project normalized burden	Project time to delivered value per unit of delivery-episode coupling load: $\tau_{2,p} \,/\, c_{e_p^*}$
$t_{0,p}$	Project start	Timestamp of earliest episode initiation: $\min_{e \in E_p} t_{0,e}$
n_p	Project partners	Number of actors contributing data in e_p^*
D_p	Domain set	Domains involved in a project
L_p	Layer set	Institutional layers involved in a project
$h_{d,p}$	Domain heterogeneity	Heterogeneity of domains involved in a project
$h_{\ell,p}$	Layer heterogeneity	Heterogeneity of layers involved in a project
$\tau_{2,p}$	Project time to delivered value	Time from the earliest episode initiation to data-product delivery: $t_{2,e_p^*} - \min_{e \in E_p} t_{0,e}$

Projects are stratified by architecture $arch_p$ and domain steward presence $stew_p$: **custodial** ($arch_p = 0$, $stew_p = 0$), **federated without steward** ($arch_p = 1$, $stew_p = 0$), and **federated with steward** ($arch_p = 1$, $stew_p = 1$).

2.1.4 Ecosystem layer

An *ecosystem* aggregates all projects within a shared geographic and institutional context, enabling cross-site comparison. Ecosystem-layer variables are simply summary statistics of

project-layer variables, such as the project normalized burden b_p.

2.2 Case selection

The governance projects analyzed here are drawn from IDS partnerships in which the authors have been directly involved as technical or governance participants. This convenience sample is intentional: retrospective reconstruction of governance indicators requires access to operational artifacts (agreements, emails, minutes) that are rarely available to outside researchers. Direct involvement provides both artifact access and the contextual knowledge necessary to delineate episode boundaries and classify events.

The resulting dataset contains nine projects comprising 30 governance episodes across three ecosystems (A, B, C), four domains (Education, Justice, Health, Human Services), and three institutional layers (Local, Regional, State), covering approximately 600,000 individuals.

Seven of the nine projects are associated with Ecosystem A, which provides intra-ecosystem variation across architectural strata (custodial and federated) but introduces potential confounding from shared institutional context. The projects outside Ecosystem A—P4 (Ecosystem B, Justice-domain, State-layer) and P8 (Ecosystem C, Health-domain, Region-layer)—provide inter-ecosystem comparisons.

3 Results

Table 5: **Project composition by domain and institutional layer.**

Project	**Eco.**	n_p	**Partner Domains**	**Partner Layers**
P1	A	3	1 Education, 1 Health, 1 Human Services	3 Local
P2	A	—	—	—
P3	A	9	3 Education, 1 Health, 5 Human Services	8 Local, 1 Regional
P4	B	2	1 Human Services, 1 Justice	2 State
P5	A	7	7 Education	7 Regional
P6	A	4	4 Education	4 Local
P7	A	2	1 Health, 1 Human Services	1 Local, 1 Regional
P8	C	4	4 Education	4 Local
P9	A	4	3 Education, 1 Health	3 Local, 1 Regional

3.1 Ecosystem A

Table 6: **Ecosystem A: project-level summary.** "—" indicates metric not reconstructable.

Project	$arch_p$	$stew_p$	n_p	$c_{e_p^*}$	$\lvert E_p \rvert$	Stall frac.	$\tau_{2,p}$ (days)	b_p
P1	0	0	3	5	10	0.40	624	124.80
P2	0	0	—	—	2	1.00	—	—
P3	1	1	9	16.90	1	0.00	87	5.15
P5	1	1	7	7.00	2	0.00	235	33.57
P6	0	0	4	4.00	1	0.00	130	32.50
P7	1	0	2	4.50	4	0.00	123	27.33
P9	1	0	4	7.56	5	0.00	114	15.08

Custodial architecture stratum (P1, P2, P6). P1 has 10 episodes: 6 committed and 4 stalled. P2 exhibits complete metric loss: 2 episodes initiated, no reconstructable indicators.

Federated architecture stratum (P3, P5, P7, P9). All four projects achieve $\tau_{2,p}$ with zero stalls. P3, P5, and P9 operate with the highest coupling loads $c_{e_p^*}$ in the dataset (16.90, 7.00, and 7.56). P3 and P5 operate with domain stewards ($stew_p = 1$).

3.2 Ecosystem B and Ecosystem C

Table 7: **Ecosystem B and Ecosystem C: project-level summary.**

Project	**Ecosystem**	$arch_p$	$stew_p$	n_p	$c_{e_p^*}$	$\lvert E_p \rvert$	Stall frac.	$\tau_{2,p}$ (days)	b_p
P4	B	1	0	2	3	3	0.00	317	105.67
P8	C	1	0	4	4	2	0.00	365	91.25

P4 (Justice-domain, State-layer) and P8 (Health-domain, Region-layer) both have PET-enabled federated architectures ($arch_p = 1$) without domain stewards ($stew_p = 0$). Both achieve zero stalls.

3.3 Cross-ecosystem summary

Table 8: **Cross-ecosystem project-level summary.**

Project	**Ecosystem**	$arch_p$	$stew_p$	$\tau_{2,p}$ (days)	$c_{e_p^*}$	b_p	n_p
P1	A	0	0	624	5.00	124.80	3
P2	A	0	0	—	—	—	—
P3	A	1	1	87	16.90	5.15	9
P4	B	1	0	317	3	105.67	3
P5	A	1	1	235	7	33.57	7
P6	A	0	0	130	4.00	32.50	4
P7	A	1	0	123	4.50	27.33	2
P8	C	1	0	365	4.00	91.25	4
P9	A	1	0	114	7.56	15.07	4

Table 9: **Indicator means by architectural stratum, all projects**

Stratum	$\bar{c}_{e_p^*}$	$\bar{\tau}_{2,p}$	$\sigma_{\tau_{2,p}}$	$\bar{b}_p$
Custodial ($arch_p = 0$)	4.50	377.00	349.3	78.65
Federated - steward ($arch_p = 1,\ stew_p = 0$)	4.77	229.75	130.0	59.83
Fed. + steward ($arch_p = 1,\ stew_p = 1$)	11.95	161.00	104.7	19.36
All federated ($arch_p = 1$)	7.16	206.83	116.6	46.34

Key patterns

1. All 6 stalled episodes occur within custodial projects ($arch_p = 0$), yielding a custodial stall fraction of 0.462; the federated stratum exhibits zero stalls across 17 episodes despite higher mean coupling loads (7.16 vs. 4.50).

2. Mean normalized time-to-value burden b_p is 78.65 in the custodial stratum versus 46.34 federated; within the federated stratum, steward-present projects achieve 19.36 versus 59.83 without stewards.

3. The highest and lowest project time-to-value in the dataset are both in Ecosystem A: the custodial project P1 reaches 624 days, while the steward-present federated project P3 achieves 87 days at the highest coupling load in the dataset (16.90). The two steward-present projects P3 and P5 handle the largest partner networks ($n_p = 9$ and 7) at high coupling loads (16.90 and 7.00), yet achieve low normalized burdens among their respective strata (5.15 and 33.57).

4 Analysis

In our experience the three patterns reported above are not adequately explained by any single-dimensional model of governance adequacy—whether framed as capacity, legitimacy, or alignment. Instead, the patterns suggest that governance adequacy comprises two independent dimensions:

Structural resolution is the subset of request types a governance boundary can distinguish. Structural resolution is discrete and combinatorial—a matter of how the boundary discriminates the requests it receives. Changing structural resolution requires structural changes to the IDS, such as adding evaluators, deploying PETs, or restructuring review procedures.

Functional capacity is how rapidly a governance boundary produces authorizations for each type of request it receives. Functional capacity is continuous and temporal—processing rate and completion probability. Improving functional capacity means improving staffing, expertise, templates, or facilitation.

4.1 Implications

Resolution constrains governance throughput. For a given request type recognized by a governance boundary, functional capacity determines the speed of request processing; but the set of types the boundary can resolve is fixed by the boundary configuration. No increase in functional capacity can compensate for a boundary that conflates structurally distinct request types. Conversely, refining the boundary's configuration does not automatically improve throughput.

The coarsening cascade. When a boundary faces capacity pressure, a common adaptive response is to coarsen request types. This trades structural resolution for functional headroom, with cascading effects: capacity pressure → coarsening → conflation → misrouted governance → friction attributed to relationships, when the root cause is structural.

Joint resolution–capacity operations. Operations that *refine* the boundary's configuration—adding a domain steward, deploying a PET—can improve both dimensions simultaneously. For example, a domain steward separates conflated request types into multiple distinct types *and* brings domain-specific expertise that accelerates processing of those types.

Cross-boundary couplings and scale As an ecosystem adds partners and domains, cross-boundary couplings requiring governance reconciliation grow faster than the number of partners alone, since each new cross-domain relationship can introduce structurally incompatible authorization requirements.

Structural changes address this: modularization reduces the volume of couplings routed through any single boundary; PET-enabled verification reduces per-coupling governance

effort; governed interfaces and domain stewards distribute commit capacity across multiple loci. Each change affects the boundary's structural configuration as well as the boundary's functional capacity.

4.2 Predictions

Our two-dimensional model of governance adequacy makes a number of empirically-verifiable predictions:

1. **Heterogeneity-sensitive commit delay.** $\tau_{1,p}$ and $\mathrm{Var}(\tau_{1,p})$ should increase with cross-domain expansion under single-boundary governance.
2. **Pre-commit stall as saturation signature.** Saturation should manifest as increased stall fraction and longer cycle times.
3. **Architecture changes normalized burden.** PET-enabled architectures should reduce normalized burden b_p.
4. **Friction redistribution, not elimination.** Governance friction reduction should shift the binding constraint to data product delivery and use.
5. **Domain stewardship attenuates normalized burden.** Projects with stewards ($stew_p = 1$) should exhibit lower normalized burden b_p, particularly at higher coupling loads c_p.
6. **Episodes with complete metric loss.** Episodes with complete metric loss are consistent with a boundary whose configuration is so coarse that no appropriate output can be produced.
7. **Stewardship as joint operation.** The steward effect should exceed what functional capacity gain alone predicts, because stewards both refine the boundary's configuration (separating conflated request types) *and* add evaluator bandwidth.

4.2.1 Falsification

If, on the other hand, our two-dimensional model does not reflect reality, then:

- Coupling load does not predict stall or latency.
- Stall rates are comparable under federated and custodial architectures.
- Normalized burden does not differ between architectural strata, or is higher under federation.
- Data product delivery and use is not a binding constraint under federated architectures.
- Domain stewardship does not attenuate normalized burden.
- The steward effect is fully explained by staffing without invoking structural resolution refinement.

4.3 Evaluating the predictions

The data we have gathered are consistent with our predictions and do not trigger any of the falsification conditions listed above.

1. **Stall concentration.** The concentration of all stalls in the custodial stratum—at coupling loads below the federated stratum's—indicates that stall is a boundary-design phenomenon rather than a scale phenomenon. The custodial boundary's inability to produce authorizations for certain request types is not explained by overload alone.

2. **Normalized burden.** The drop in normalized burden from custodial to federated-with-steward strata (78.65 to 19.36) is consistent with the prediction that architectural and governance design correlate with time-to-value. The steward-present project P3 achieves the lowest normalized burden (5.15) at the highest delivery-episode coupling load in the dataset ($c_{e_p^*} = 16.90$), which is the opposite of what a pure capacity-scaling explanation would predict. The federated-without-steward stratum (59.83) shows a more modest reduction from custodial, and is inflated by two ecosystem-seeding pilots (P4, P8) whose high burdens reflect institutional bootstrapping rather than architectural inadequacy.

3. **Friction redistribution.** The shift from governance-dominated to delivery-dominated time-to-value under federation indicates that governance friction is not eliminated but redistributed: reducing the authorization bottleneck reveals the delivery bottleneck that was previously masked.

4. **Domain stewardship.** The steward-present projects (P3, P5) managed the most heterogeneous partnerships. Both operated through backbone organizations that served as natural interpreters of domain-specific compliance language, suggesting that stewardship works in part by distributing interpretive authority.

5. **P2 as candidate resolution-floor case.** P2 exhibits complete metric loss despite engagement—consistent with a boundary whose configuration is too coarse to produce the specific artifacts that durable authorization requires.

6. **Stewardship as joint operation.** The steward effect (roughly four-fold normalized burden reduction relative to the custodial stratum, and three-fold relative to federated-without-steward) exceeds what functional capacity gain alone would predict. Domain stewards perform a joint operation: separating conflated categories (e.g., distinguishing "linkage with plaintext data" from "linkage with PET-mediated data" as distinct request types) *and* increasing evaluator bandwidth. The effect is most visible in P3, where stewardship combined with PET architecture and ecosystem maturity produces the lowest burden at the highest coupling load.

4.4 Cross-ecosystem analysis

The two non-Ecosystem A projects (P4 and P8), which operate in different domains and institutional layers from the Ecosystem A majority (Section 2.2), achieve zero stalls under federated architecture. However, their normalized burdens (105.67 and 91.25) place them among the highest in the dataset.

This result is consistent with both projects' status as ecosystem-seeding pilots: each was the first IDS project in its community, requiring construction of institutional infrastructure—first agreements, first trust relationships, first governance precedents—from scratch. P1, the custodial pilot in Ecosystem A, exhibits similarly elevated burden (124.80).

Meaningful burden comparison across architectural strata likely requires controlling for ecosystem maturity. Within the mature Ecosystem A context, where institutional precedent is established, the architectural and stewardship effects on burden are clearer.

5 Discussion

Our methods are diagnostic, not prescriptive, and our diagnostic analysis is limited to the data we have recorded in our own IDS projects. With such a small sample (nine projects and 30 episodes), summaries are sensitive to individual projects. Seven of the nine projects are in Ecosystem A, introducing potential confounding from shared institutional context. P2's complete metric loss prevents comparison with completed episodes.

5.1 Alternative explanations

Several alternative explanations for the observed data merit consideration:

Institutional learning. In our experience custodial projects often precede federated projects in the same ecosystem. The observed differences could partially reflect accumulated experience rather than architectural changes. The non-Ecosystem A projects (P4, P8) show zero stalls consistent with the federated pattern, but their elevated normalized burdens—comparable to the custodial pilot P1—suggest that ecosystem maturity, not just architecture, contributes to burden reduction. Within Ecosystem A, where institutional precedent exists, the architectural effect is clearer.

Selection effects. Projects that adopted federated architecture may have done so precisely because they anticipated governance challenges, selecting into the architecture that the data suggest works better. This cannot be ruled out with the available data, though it would not explain why steward-present projects outperform steward-absent federated projects at higher coupling loads.

Personnel and leadership. Differences in individual competence or leadership quality could account for some variation. However, the pattern across architectural strata—where the architecture varies while the institutional ecosystem is held roughly constant—suggests that organizational design, not personnel, is the primary factor.

5.2 Implications for practice

The concentration of all stalled episodes in the custodial stratum—at coupling loads *below* those handled under federation—challenges the common assumption that governance breakdown reflects deficits of trust, leadership, or alignment. Even a high-throughput boundary cannot absorb growing reconciliation burden when its configuration is too coarse for the request types it faces. Instead, persistent governance delay, rising latency, and increasing variance should be evaluated as potential signals of governance saturation *before* attributing them to misalignment or insufficient effort. When saturation is misdiagnosed as a throughput problem, the typical response—more staffing, more facilitation—addresses operating parameters but not the boundary's structural configuration.

Three practical implications follow:

1. **Centralizing governance concentrates friction at a single boundary.** Ecosystems expanding across domains should anticipate nonlinear growth in governance burden.

2. **Process improvement addresses functional capacity, not structural resolution.** Staffing and templates cannot resolve conflation of request types.

3. **Scalable governance requires structural change.** Potential changes include reducing coupling growth (modularization), increasing verification-based compliance (PET-enabled processing), or distributing commit capacity across multiple boundaries (governed interfaces, domain stewards).

5.3 Friction redistribution and governance design

Burden reduction at substantially higher coupling loads suggests that scalable governance benefits not only from boundary redesign but from *complementary institutional design* that distributes interpretive authority along domain lines. Custodial architecture effectively demands one authority process all authorizations at a coarse resolution. Domain stewards distribute that interpretive work: each steward maintains their domain's necessary distinctions while coordinating through shared protocols. As a result, the steward-present projects in our dataset achieve the lowest normalized burdens at the highest coupling loads. In both projects, the steward organization possessed pre-existing authority, trust relationships, and compliance infrastructure within its domain.

In our practice we recommend IDS projects identify natural domain stewards *before* selecting technologies. Practitioners building or expanding IDS partnerships should prioritize identifying organizations that can serve as domain stewards—backbone organizations, collective impact intermediaries, or agencies with established cross-agency convening roles—as a governance design decision, not merely a staffing decision.

This pattern—distributed custody with coordinated governance—is consistent with the composable data mesh paradigm [14], in which domain teams own their data products and

participate in a federated governance layer. However, our results suggest that the governance benefits of composability are not automatic; they depend on whether the governance boundary's structural resolution matches the heterogeneity of the domains being composed.

5.4 Future work

We plan to continue applying the methods described here in ongoing and new IDS projects, with the long-term goal of replicating and refining our model across ecosystems with varying parameters (architecture, stewardship). We also plan to develop tools to capture event- and episode-level data as they arise rather than retrospectively, enabling running project analysis. Finally, we intend to develop a complementary model for diagnosing bottlenecks in data pipeline construction, identity resolution, and analytic capacity—bottlenecks that can be addressed through composable, domain-owned infrastructure designs.

6 Conclusion

In this paper we introduced a set of minimal, reconstructable indicators and a layered analytical model for diagnosing governance saturation in Integrated Data Systems. We applied the model retrospectively to nine projects in three ecosystems. In the course of our analysis we developed a new, two-dimensional model of governance adequacy.

Persistent governance delays in cross-domain and -layer IDS projects should be considered as a potential signal of structural mismatch before they are attributed to deficits of trust, leadership, or effort. When the structural resolution of a governance boundary configuration is too coarse for the request types the boundary faces, capacity increases alone are unlikely to improve governance throughput.

Scalable governance designs must address both structural resolution and functional capacity and should consider structural changes such as modularization, PETs, and domain stewardship. The most important design question is not whether governance friction can be eliminated, but whether it is distributed to boundaries with appropriate resolution and capacity. Finally, practitioners should anticipate that alleviating governance saturation may reveal previously-obscured delivery bottlenecks.

7 Acknowledgments

The author thanks Dr. Minor Gordon for detailed review and substantive contributions to this manuscript. The author also acknowledges the IDS partner organizations whose governance processes generated the operational artifacts analyzed in this study. This research was not supported by external funding.

8 Statement on Conflicts of Interest

The author is founder and principal of Asemio, which provides technical infrastructure and consulting for IDS partnerships, including several of the projects analyzed in this study. Asemio's involvement provided the artifact access required for retrospective analysis (Section 2.2) but also constitutes a potential commercial interest in the findings.

9 Ethics Statement

This study analyzed operational artifacts (signed agreements, emails, committee minutes, provisioning logs) generated in the routine course of IDS governance processes. No individual-level data were accessed, analyzed, or reported; all indicators are derived from institutional-level governance events. The study does not constitute human subjects research and was not submitted for institutional review board (IRB) approval. Population counts reported in the case descriptions are approximate figures provided for context and do not reflect access to individual records.

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10 Abbreviations

FERPA	Family Educational Rights and Privacy Act
HIPAA	Health Insurance Portability and Accountability Act
IDS	Integrated Data System
PET	Privacy-Enhancing Technology

11 Data Availability Statement

The operational artifacts underlying this analysis (signed agreements, emails, committee minutes, provisioning logs) contain institutionally sensitive information belonging to the partner organizations involved in each IDS project. These artifacts cannot be made publicly available without the consent of all participating institutions. The derived project-level and episode-level indicators reported in Tables 6–9 constitute the complete set of metrics used in the analysis. Researchers seeking additional detail may contact the author.

12 AI Disclosure Statement

The author used AI-assisted tools (Anthropic Claude Opus 4.6) during the preparation of this manuscript for editorial feedback, error checking, and LaTeX formatting. All analytical or methodological decisions, variables and definitions, theoretical interpretations, and substantive content are the author's own. The author reviewed and takes full responsibility for the final text.